

AI Interviewer

AI-Powered Interview Preparation Web App

Design Document

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Prepared by: Christian Hernandez

1. Executive Summary

PrepAI is a web-based AI-powered interview preparation platform designed to help software engineering candidates practice for technical interviews with realistic simulations. The platform runs entirely in the browser, combining cutting-edge AI technologies including real-time video avatars, speech analysis, and intelligent question generation.

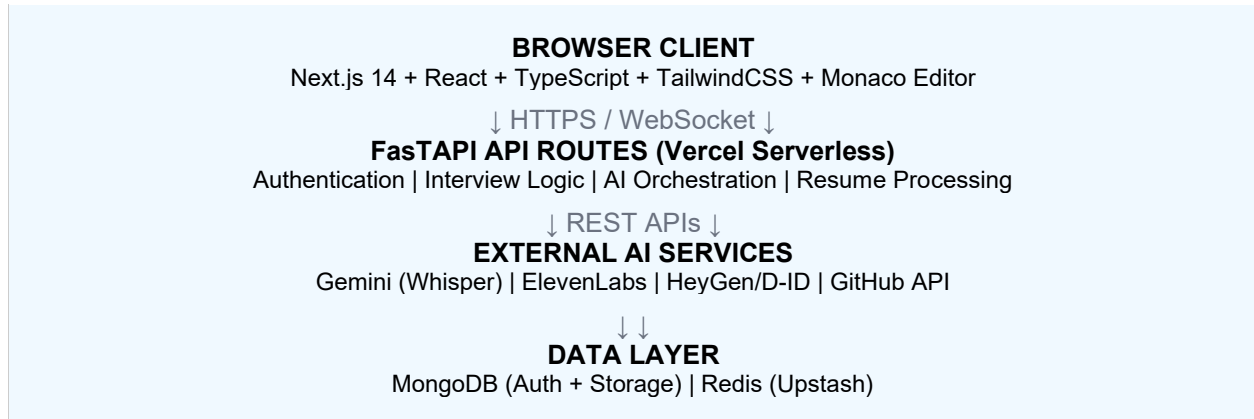
Key Features:

- Company-specific mock interviews with curated LeetCode questions
- AI-powered video interviewer with realistic avatar and voice
- Resume-based technical questioning about projects and experience
- Behavioral interview practice with STAR method guidance
- Comprehensive feedback on delivery, filler words, pacing, and technical accuracy
- AI resume builder that transforms GitHub projects into polished resumes

2. System Architecture

2.1 High-Level Architecture

The system follows a modern full-stack web architecture with Next.js handling both frontend and API routes, deployed serverless on Vercel with Supabase as the backend-as-a-service.



2.2 Technology Stack

Layer	Technology	Purpose
Frontend	Next.js 14, React 18, TypeScript, Tailwind CSS	SSR, responsive UI, type safety
Code Editor	VSCODE	VS Code-like coding environment
Backend	FastAPI Routes (App Router)	Serverless API endpoints
Database	PostgreSQL	Relational data, RLS security
Auth	PostgreSQL Auth	OAuth (Google, GitHub), JWT
Storage	PostgreSQL Storage	Videos, resumes, uploads
Caching	TBD	Rate limiting, session cache
AI - LLM	Gemini	Question generation, feedback, resume writing
AI - Speech	TBD	Speech-to-text transcription
AI - Voice	TBD	Text-to-speech for interviewer
AI - Avatar	HeyGen / D-ID API	Realistic video avatar generation
Deployment	Vercel (cause its free)	Serverless, edge functions, CI/CD
Payments	TBD (based on legality of data stuff)	Subscriptions, checkout

3. Core Features

3.1 Company-Specific Mock Interviews

Users select from a curated list of top tech companies, each with company-specific question pools sourced from LeetCode tagged questions and verified interview experiences.

- **Supported Companies:** Google, Meta, Amazon, Apple, Microsoft, Netflix, Tesla, NVIDIA, Stripe, Uber, Airbnb, LinkedIn
- **Question Categories:** Arrays, Strings, Trees, Graphs, (rest is things like Dynamic Programming, System Design but wanna keep it simple)
- **Difficulty Levels:** Easy, Medium, Hard (Depends on company)

3.2 AI Video Interviewer

A photorealistic AI avatar conducts the interview in real-time within the browser, creating an immersive video interview experience.* PROB START AS CHATBOT

- **Avatar:** HeyGen/D-ID streaming API for realistic lip-synced video *CLAUDE RECOMMENDED
- **Voice:** ElevenLabs for natural speech synthesis
- **Listening:** Whisper API for accurate speech-to-text
- **Video:** getUserMedia API for candidate webcam capture

3.3 Resume-Based Technical Questions

The AI analyzes uploaded resumes to generate personalized deep-dive questions about listed projects and technologies.

- **Parsing:** PDF.js extraction + GPT-4 structured parsing
- **Analysis:** Extract projects, skills, technologies, achievements
- **Questions:** "Tell me about the challenges you faced building [Project X]" *hardcoded at first most likely

3.4 Behavioral Interview Module

- **Categories:** Leadership, Conflict, Teamwork, Problem-Solving, Failure/Learning
- **STAR Framework:** Situation, Task, Action, Result coaching
- **Company-Specific:** Amazon Leadership Principles, Google Googliness, etc.

3.5 Interview Feedback & Analytics

Post-interview analysis provides actionable insights:

Metric	Details
Speech Analysis	Filler words (um, uh, like), pace (WPM), volume, clarity score
Technical Accuracy	Code correctness, complexity analysis, edge cases, optimizations
Communication	Explanation clarity, thought verbalization, question clarification
Time Management	Time per question, pacing, rushing vs over-explaining
Overall Score	Weighted composite with percentile ranking

3.6 AI Resume Builder

Transform raw information into polished, ATS-optimized resumes:

- **GitHub Import:** OAuth to pull repos, READMEs, languages, contribution stats
- **Natural Language:** Describe experience conversationally, AI structures it
- **Enhancement:** Action verbs, quantified achievements, ATS keywords
- **Export:** PDF download or shareable web link

4. Database Schema (Supabase/PostgreSQL)

***CLAUDE MADE RECOMMENDED THIS**

Table	Columns	Relations
users	id, email, name, avatar_url, subscription_tier, created_at	1:N interviews
companies	id, name, slug, logo_url, description, interview_style	1:N questions
questions	id, company_id, category, difficulty, text, solution, hints	N:1 company
interviews	id, user_id, company_id, type, status, recording_url, timestamps	N:1 user
responses	id, interview_id, question_id, transcript, code, score	N:1 interview
feedback	id, interview_id, speech_metrics, tech_score, comm_score, overall	1:1 interview
resumes	id, user_id, content_json, pdf_url, version, created_at	N:1 user

5. API Routes (Fast API App Router)

Method	Route	Description
*	/api/auth/[...nextauth]	NextAuth.js handlers
GET	/api/companies	List all companies
GET	/api/companies/[slug]/questions	Questions for company
POST	/api/interviews	Create interview session
GET	/api/interviews/[id]	Get interview state
POST	/api/interviews/[id]/respond	Submit answer
POST	/api/interviews/[id]/end	End and process feedback
POST	/api/ai/transcribe	Whisper speech-to-text
POST	/api/ai/avatar	Generate avatar response
POST	/api/ai/analyze	GPT feedback analysis
POST	/api/resume/parse	Parse uploaded resume
POST	/api/resume/generate	Generate from inputs
GET	/api/github/repos	Fetch user repos
POST	/api/stripe/checkout	Create checkout session
POST	/api/stripe/webhook	Handle Stripe events

6. Frontend Pages

Route	Description
/	Landing page - hero, features, pricing, testimonials
/login	OAuth login (Google, GitHub buttons)
/dashboard	User home - stats, recent interviews, quick actions
/companies	Browse companies with cards, search, filters
/interview/setup	Configure interview - type, resume upload, settings
/interview/[id]	Live interview - avatar video, code editor, timer
/interview/[id]/feedback	Results - scores, transcript, improvement tips
/resume	Resume builder - input methods, preview
/resume/[id]/edit	Edit resume with live preview
/history	All past interviews with filters
/settings	Account, subscription, preferences
/pricing	Plans comparison, Stripe checkout

7. User Flows

7.1 Interview Flow

1. **Login:** OAuth with Google or GitHub
2. **Select Company:** Browse cards, click to select
3. **Choose Type:** Technical / Behavioral / Full
4. **Upload Resume:** Optional drag-drop for personalized Qs
5. **Permissions:** Grant camera/mic access
6. **Interview:** AI avatar asks questions, user responds
7. **Code:** Monaco editor for technical questions
8. **Feedback:** View scores, transcript, tips

7.2 Resume Builder Flow

9. **Choose Input:** GitHub / Manual / Upload existing
10. **GitHub Auth:** One-click OAuth
11. **Select Projects:** Checkboxes for repos to include
12. **Add Experience:** Natural language text input
13. **Generate:** AI processes and structures
14. **Preview:** Live editable preview
15. **Export:** Download PDF or share link

9. Development Timeline

Phase	Deliverables
Week 1-3(after charter)	
Week 4-7	
Week 8-10	
Week 11-13	
Week 14-15	

10. Conclusion

PrepAI is a focused web application delivering comprehensive interview preparation entirely in the browser. By leveraging Next.js, Supabase, and modern AI APIs (OpenAI, ElevenLabs, HeyGen), the platform provides realistic mock interviews, intelligent feedback, and automated resume building without requiring downloads or installations.

The serverless architecture on Vercel ensures cost-effective scaling, while the modular Next.js structure enables rapid iteration. With a clear 15-week timeline and straightforward Stripe monetization, PrepAI is positioned to help software engineering candidates ace their interviews.

Backend structure

backend/

controllers/

interview_controller.py

ai_controller.py

resume_controller.py

feedback_controller.py

Services/ separate system prompts

llm_service.py

interview_service.py

resume_service.py

feedback_service.py

Models/ -- Abel

user.py

interview.py

Resume.py

PRODUCT BACKLOG for SPRINT 2:

1) Designing and implementing initial user interface
- 18 Hours

2) Implementing APIs and Backend routes – 40Hours

3) Building a relational database – 15 Hours

Sprint Backlog for 1) Designing and implementing initial UI

A) Interview Input form – 5 Hours

B) Basic Styling and layout – 8 hours

C) Connecting Frontend to backend through our FastAPI routes – 3 Hours

Sprint Backlog for 2) implementing APIs and Backend routes

A) Setting up the FastAPI backend structure - 4 Hours

B) connect routes to the database to store user Input - 6 Hours

C) AI interview plan generation via gemini - 10 Hours

D) Real Time text to text interview via websockets or basic rest if time is an issue – 12 Hours

E)Basic post interview feedback Generation – 8 Hours

Sprint Backlog for 3) Building a relational database

A) Designing Database Schemas – 3 Hours

B) setup PostgreSQL environment - 1 Hour

C) Implement tables and relation ships - 5 Hours

D) Connect backend to the database – 6 Hours

DATABASE:

SPARK DATAFRAME – how to create tables, etc.

Databricks -